MediAI Care+: AI Chatbot & LLM Integration Report

Groq Cloud LLM (llama-3.3-70b-versatile), 3-Tier Failover & Clinical Decision Support

Module Focus:	AI Chatbot & Groq LLM Cloud Integration
LLM Model:	llama-3.3-70b-versatile (Groq Cloud API)
Backend Endpoint:	http://localhost:8000/api/chat (Python FastAPI)

1. Executive Overview

The MediAI Care+ Chatbot provides high-precision clinical guidance utilizing Groq's **llama-3.3-70b-versatile** engine combined with dynamic patient vitals telemetry.

2. Resilient 3-Tier Fallback Architecture

Tier Level	Target Protocol / Endpoint	Operational Function
Tier 1 (Primary)	FastAPI Backend Proxy (http://localhost:8000/api/chat)	Server proxy handling security and logging.
Tier 2 (Secondary)	Direct Groq API Endpoint (llama-3.3-70b-versatile)	Direct client-to-cloud request if local server down.
Tier 3 (Tertiary)	Dynamic Knowledge Engine (In-Memory)	Offline fallback for 80+ clinical conditions.

3. Empirical Verification

Tested & verified returning **200 OK** with high-quality clinical consultation responses.